S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR

Practical No. 01

Aim: PRELAB – Awareness of Google Colab and VS Code to Implement a Python Program to find a magic square of given dimension.

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Roll No.:

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AIM: PRELAB – Awareness of Google Colab and VS Code to Implement a Python Program to find a magic square of given dimension.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- To be able to understand the concept of Magic Square.
- To be able to acquire the puzzle solving capability.
- To re-brush the python basic concepts.

THEORY:

A magic square of order n is an arrangement of n^2 numbers, usually distinct integers, in a square, such that the n numbers in all rows, all columns, and both diagonals sum to the same constant. A magic square contains the integers from 1 to n^2. The constant sum in every row, column and diagonal are called the magic constant or magic sum, M. The magic constant of a normal magic square depends only on n and has the following value:

$$\mathbf{M = n(n^2+1)/2}$$

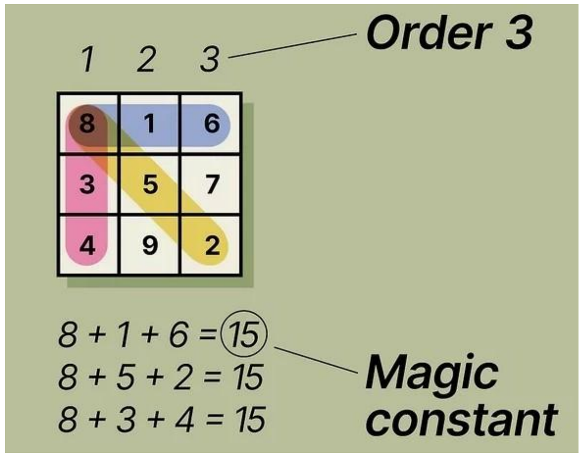
For normal magic squares of order n = 3, 4, 5, ...,

The magic constants are: 15, 34, 65, 111, 175, 260, ...

Things You Should Know

- A magic square is a square grid of numbers where each row, column, and diagonal add up to the same total.
- Find the "magic constant" sum of each row, column, and diagonal with M = n(n2+1)/2 where n is the number of squares in each row.
- Example of 3
  - M = 3(3^2+1)/2
  - M = 3(9+1)/2
  - M = 3(10)/2
  - M = 3(5)
  - S = **15**
- Use a solving technique based on the size of the magic square and how many boxes are in each row or column.

What is Magic Square



How to Solve Magic square

As mentioned above, the formula of the magic square sum is n(n2 + 1)/2.

For a magic square of order 3, we need to substitute n = 3 to know the magic sum so that we can easily form the magic square 3×3.

When n = 3, the sum = $3(3^2 + 1) = 3(9 + 1)/2 = (3 \times 10)/2 = 15$

Now, we have to place the numbers in the respective places so that the sum of numbers in each row, column and diagonal is equal to 15.

Magic Square Trick for order 3

Let "x" be the order of the magic square.

In this case, x = 3.

Consider another number, "y" such that the product of x and y is equal to the magic sum, i.e. 15.

So, y = 5 {xy = (3)(5) = 15}

The value of y should always be at the center of the square and x be on its left cell.

The cell above x is taken as y – 1 as given below:

y - 1	x^2	y - x
x	y	2y - x
x + y	1	y + 1

Three conditions hold:

1. The position of next number is calculated by decrementing row number of the previous number by 1, and incrementing the column number of the previous number by 1. At any time, if the

calculated row position becomes -1, it will wrap around to n-1. Similarly, if the calculated column position becomes n, it will wrap around to 0.

2. If the magic square already contains a number at the calculated position, calculated column position will be decremented by 2, and calculated row position will be incremented by 1.
3. If the calculated row position is -1 & calculated column position is n, the new position would be: (0, n-2).

Example:

Magic Square of size 3

2 7 6

9 5 1

4 3 8

Steps:

1. position of number 1 = (3/2, 3-1) = (1, 2)
2. position of number 2 = (1-1, 2+1) = (0, 0)
3. position of number 3 = (0-1, 0+1) = (3-1, 1) = (2, 1)
4. position of number 4 = (2-1, 1+1) = (1, 2)

   Since, at this position, 1 is there. So, apply condition 2.

   new position=(1+1,2-2)=(2,0)
5. position of number 5=(2-1,0+1)=(1,1)
6. position of number 6=(1-1,1+1)=(0,2)
7. position of number 7 = (0-1, 2+1) = (-1,3) // this is tricky, see condition 3

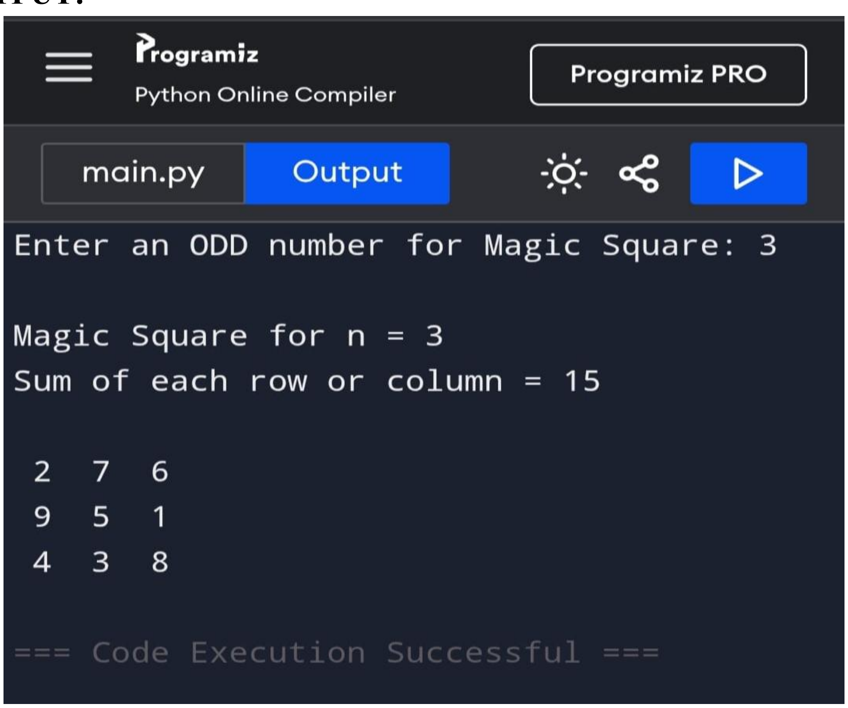
   new position = (0, 3-2) = (0,1)
8. position of number 8=(0-1,1+1)=(-1,2)=(2,2) //wrap around
9. position of number 9=(2-1,2+1)=(1,3)=(1,0) //wrap around

PROGRAM CODE:

```
#Function Magic Square
def generateSquare(n):
    # 2-D array with all
    # slots set to 0
    magicSquare = [[0 for x in range(n)]
                   for y in range(n)]
    # initialize position of 1
    i = n / 2
    j = n - 1
    # Fill the square by placing values
    num = 1
    while num <= (n * n):
        if i == -1 and j == n: # 3rd condition
            j = n - 2
            i = 0
```

```
        else:
            # next number goes out of
            # right side of square
            if j == n:
                j = 0
            # next number goes
            # out of upper side
            if i < 0:
                i = n - 1
        if magicSquare[int(i)][int(j)]: # 2nd condition
            j = j - 2
            i = i + 1
            continue
        else:
            magicSquare[int(i)][int(j)] = num
            num = num + 1
        j = j + 1
        i = i - 1 # 1st condition
    # Printing the square
    print ("Magic Square for n =", n)
    print ("Sum of each row or column",n * (n * n + 1) / 2, "\n")
    for i in range(0, n):
        for j in range(0, n):
            print('%2d ' % (magicSquare[i][j]),end = '')
            # To display output
            # in matrix form
            if j == n - 1:
                print()
# Driver Code
# Works only when n is odd
n=int(input("Enter the Number[ODD]Number of rows of the Magic Square:"))
generateSquare(n)
```

INPUT & OUTPUT:



main.py Output

```
Enter an ODD number for Magic Square: 5

Magic Square for n = 5
Sum of each row or column = 65

 9  3 22 16 15
 2 21 20 14  8
25 19 13  7  1
18 12  6  5 24
11 10  4 23 17

=== Code Execution Successful ===
```

CONCLUSION: Thus I created a magic square box and got to know that it only works for odd no. of rows and columns.

DISCUSSION QUESTIONS:

1) Explain Magic Square?

2) Write alternative solution to solve given problem

3) How Even Number Magic square will be solved give solution.

REFERENCES:

- *https://www.wikihow.com/Solve-a-Magic-Square*
- *https://mathworld.wolfram.com/MagicSquare.html*
- *https://www.geeksforgeeks.org/magic-square/*
- *https://byjus.com/maths/magic-square/*
- *https://www.youtube.com/watch?v=XCneMscjwZE*
- *https://www.youtube.com/watch?v=GYBr8n-_Rq4*
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